

From Intune Admin to Cloud Automator

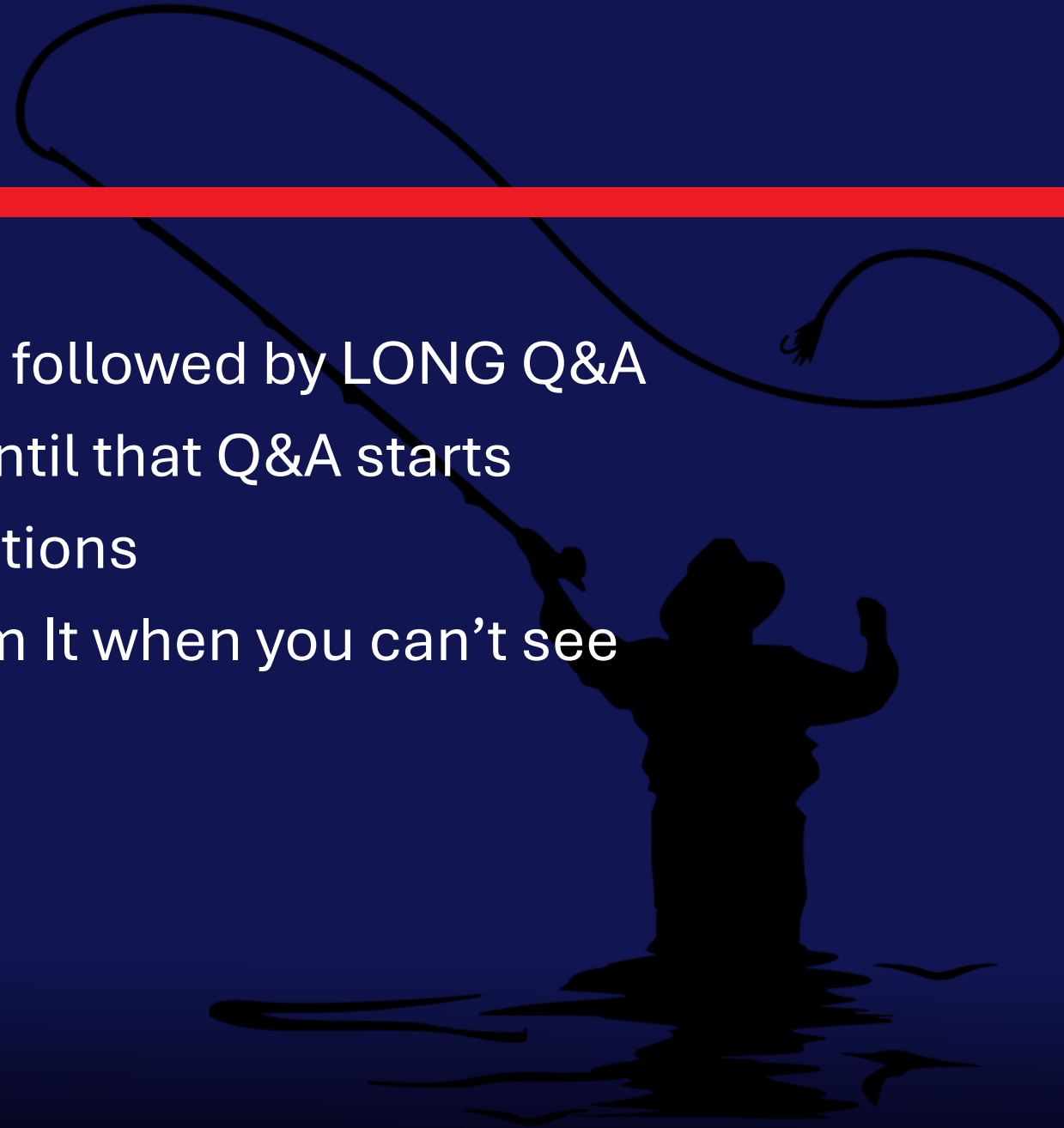
Thinking outside of the box with solutions

PREVIEW



Attention

- ◆ MMS sessions are 60-75 minutes followed by LONG Q&A
- ◆ Please hold detailed questions until that Q&A starts
- ◆ Feel free to ask clarification questions
- ◆ Remind the speakers to use Zoom It when you can't see



Speakers



Maurice Daly

Senior Security Architect



in/MauriceDaly



David Brook

CloudXP Ltd, Microsoft MVP



in/davidbrookeuc365

Can't see our slides? Can't hear? Need to repeat the question? Call us out!

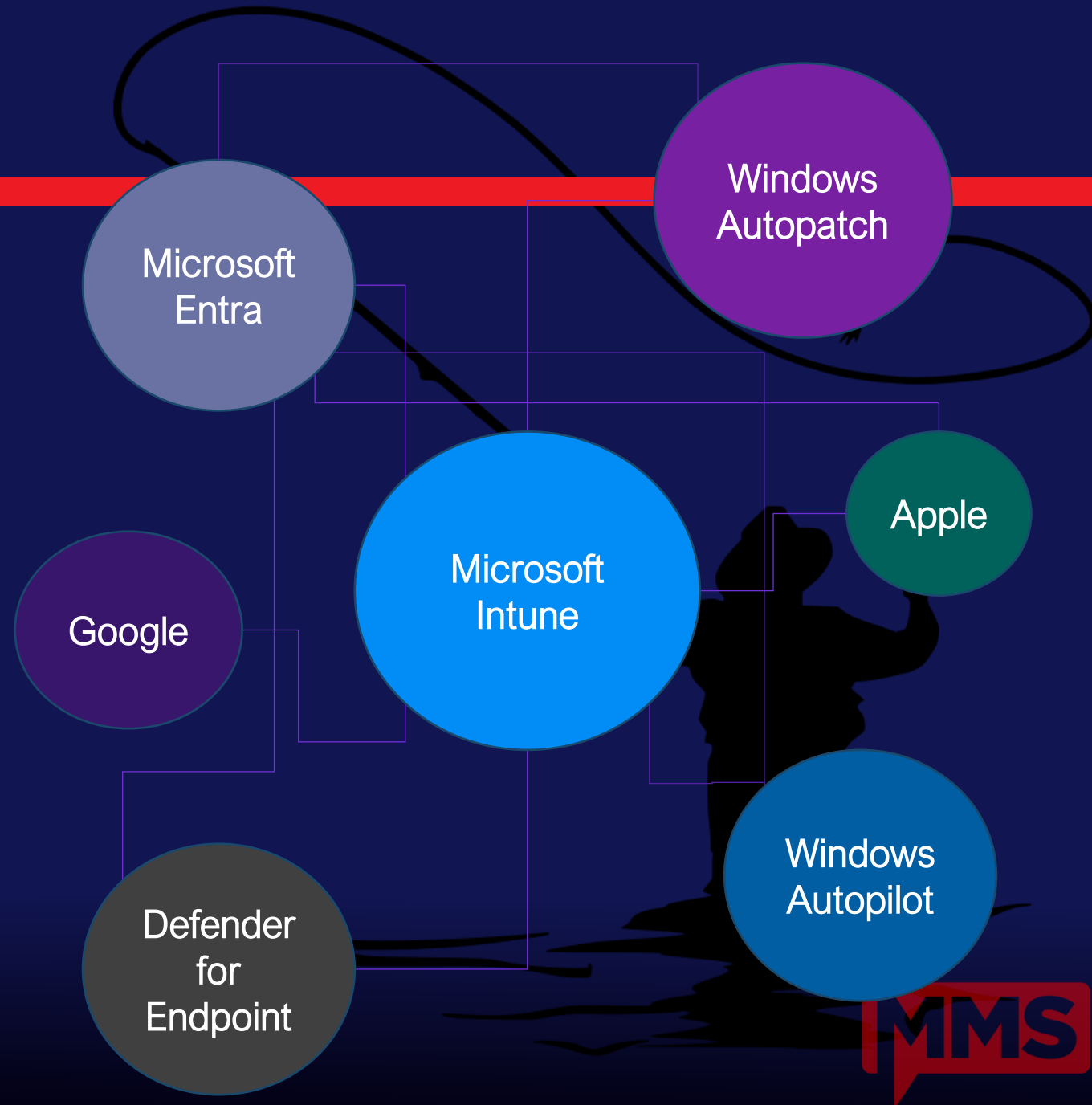
Agenda

- Understand the limits of what Microsoft Intune can do, and then go beyond them
- Learn the differences between different types of automation, including;
 - Logic Apps
 - Azure Automation
 - Function Apps
- Interacting with Restful API's to link multiple solutions together
- Get real-world examples and code snippets

Intune

When it comes to how Intune handles things like identity, anti-malware, and updates, there are many layers of spider web of integrations which help it serve your demands, including;

- Microsoft Entra
- Defender for Endpoint
- Windows Autopatch
- Windows Autopilot
- Apple Business Manager
- Google Play

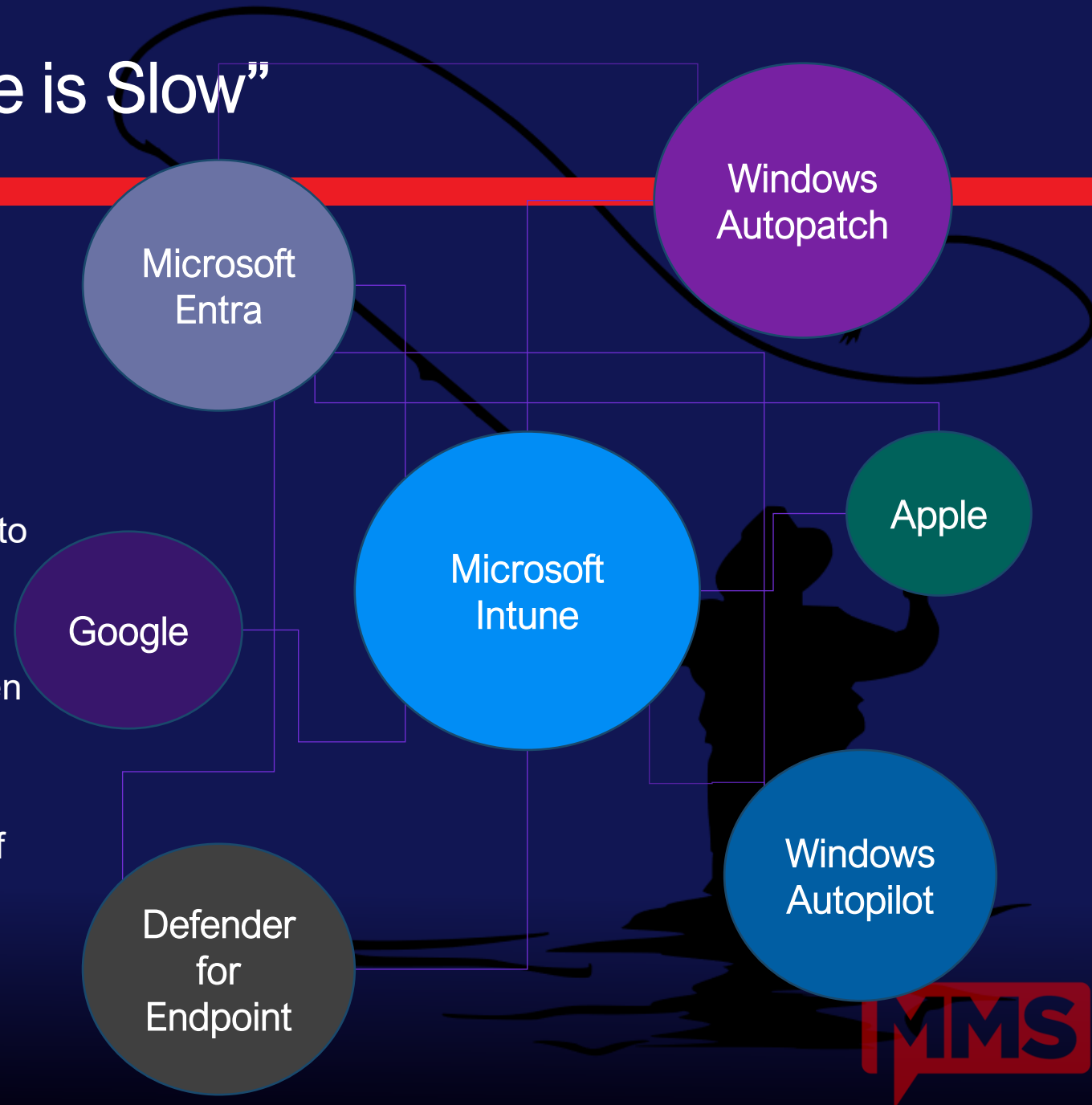


Perception of some.. “Intune is Slow”

Due to the sheer numbers when it comes to devices communicating with Intune, there can be “delays”.

This is not a failure of the product, you just have to understand you are using shared compute and resources. Therefore, things like reporting can have varying processing times, from minutes even up as far as days.

Policies on devices also have check-in periods of up to 8 hours.



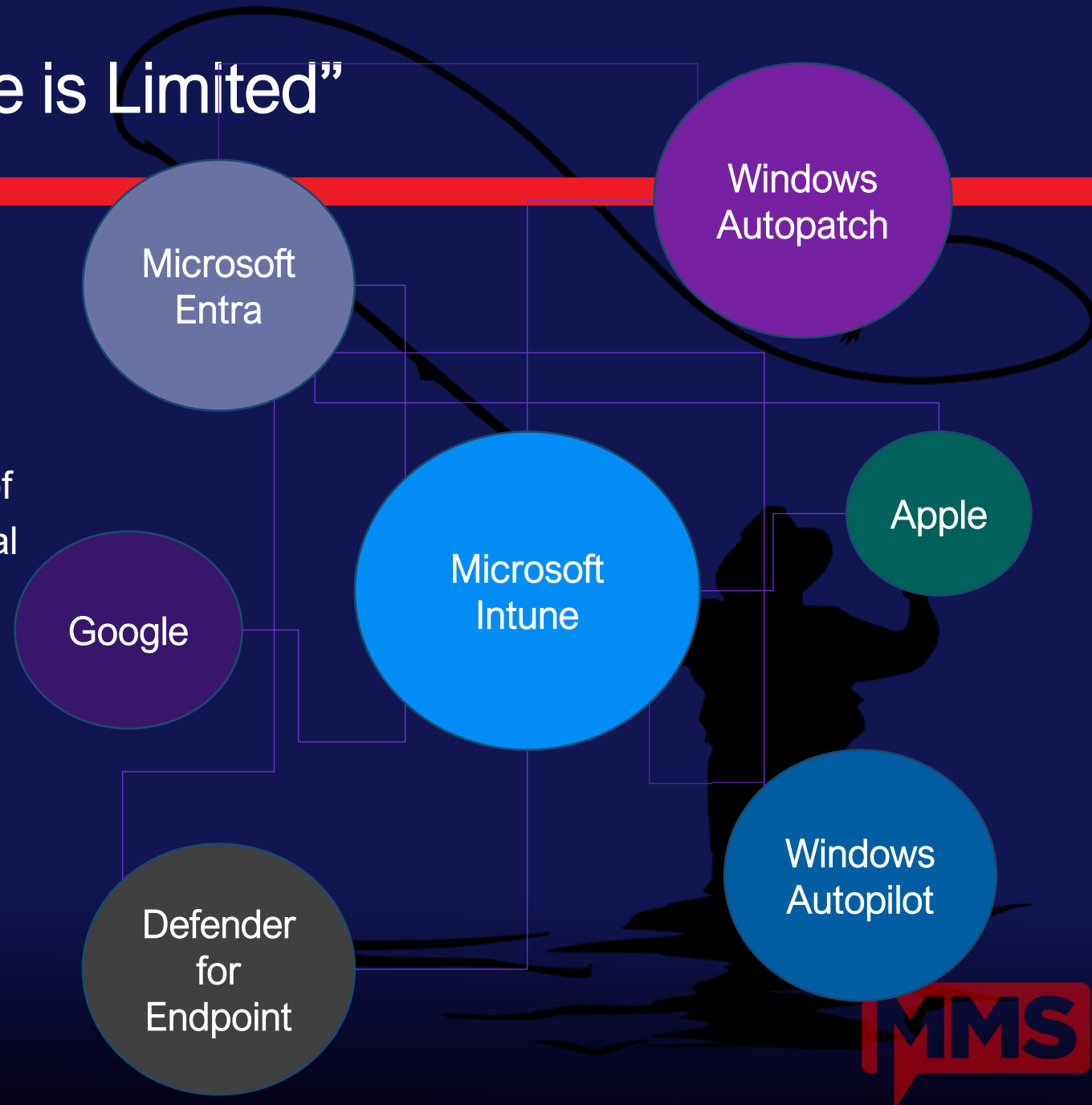
Perception of some.. “Intune is Limited”

I can't do this thing I used to do with my on-premises management

I want to name all my devices by the country of the user, but Intune only lets me have the serial number

I wish it could do... x... y.. Z...

Spoiler – It can, but you need to get creative!



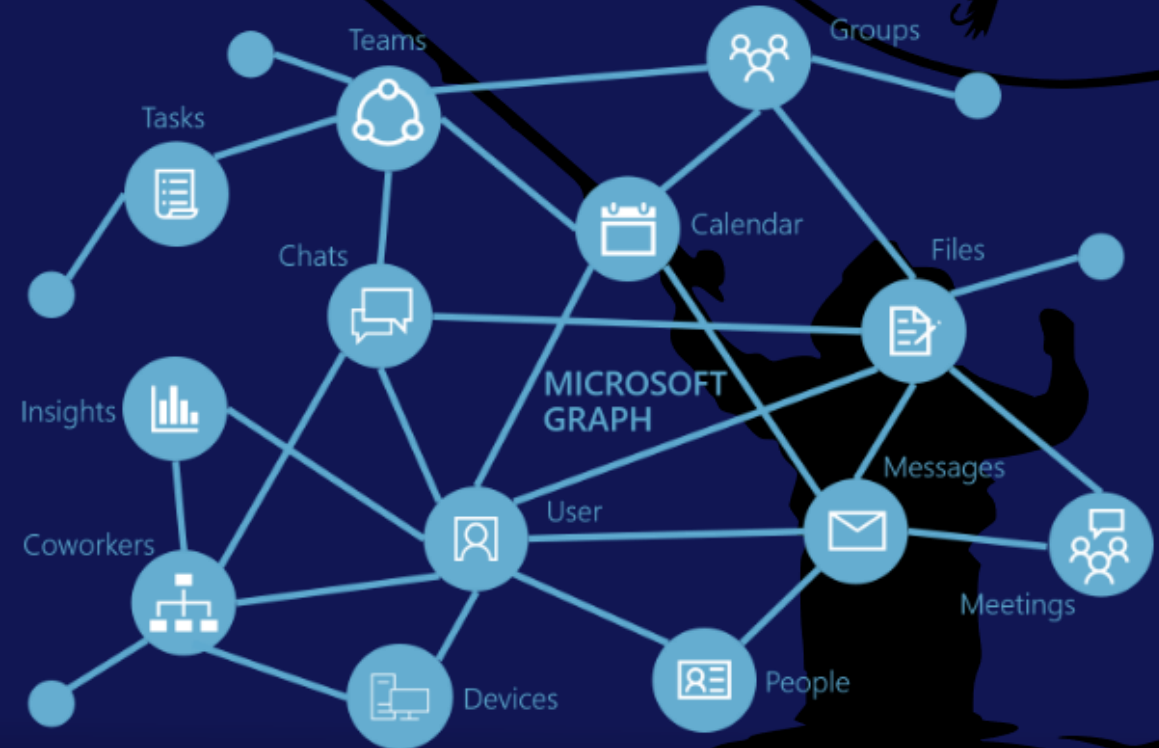
Intune has over 400 micro-services

- ◆ Before we talk about automation, let us talk about how we get the data..
- ◆ Graph API
 - ◆ Restful API

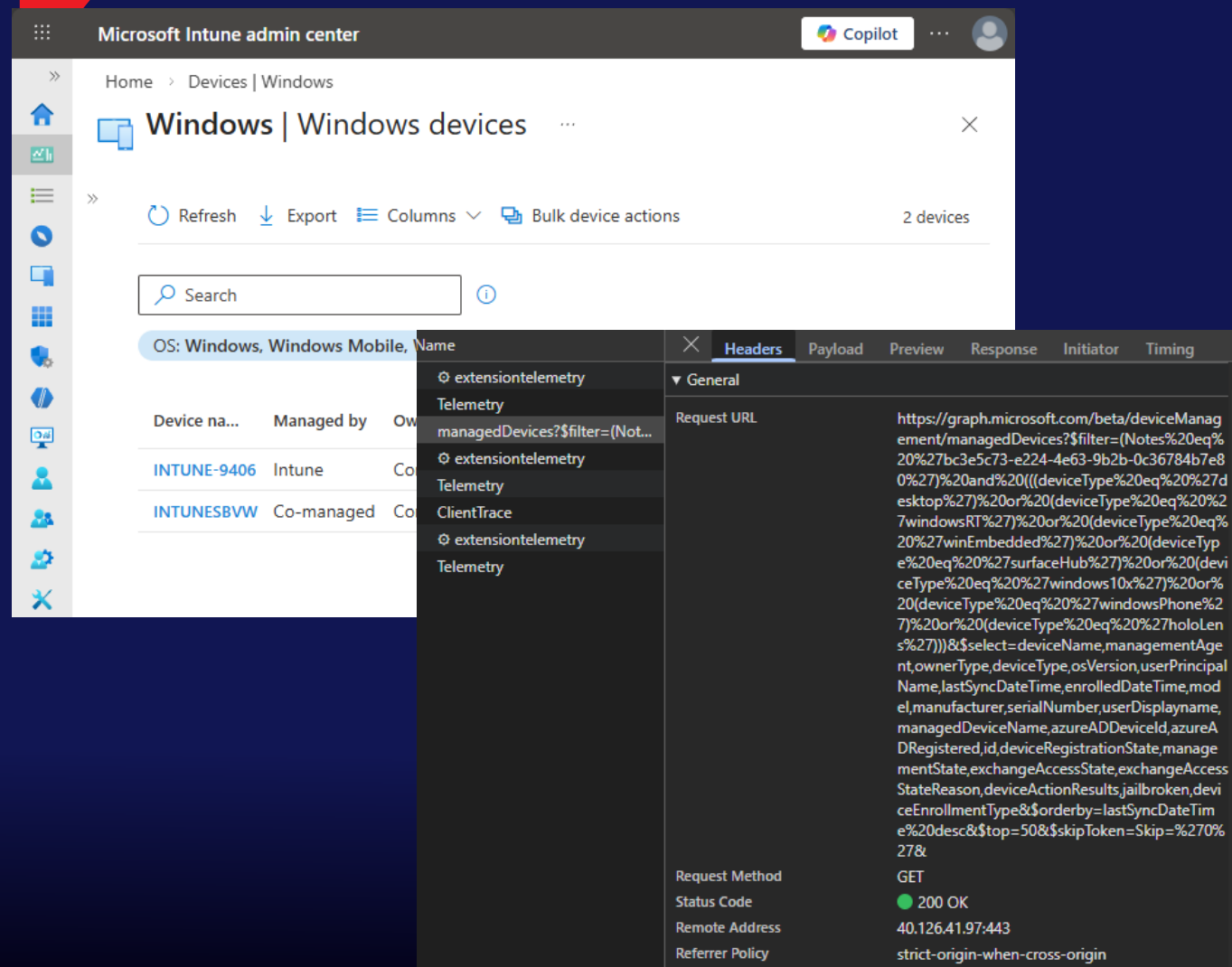
<https://graph.microsoft.com/{endpoint}/{resource}/{id}>

 - ◆ **{endpoint}** is either v1.0 or beta
 - ◆ **{resource}** is the resource to interact with
 - ◆ **{id}** is the object id of the item to interact with
- ◆ Most Microsoft services are built on Beta... Which is not actually Beta 🤔

[Working with Intune in Microsoft Graph - Microsoft Graph v1.0 | Microsoft Learn](#)



Graph 101



The screenshot shows the Microsoft Intune admin center interface. The top navigation bar includes the Microsoft Intune admin center logo and a Copilot button. The main content area is titled "Windows | Windows devices" and shows a list of devices. A search bar is present. Below the list, a detailed view of a device is shown, including a table of telemetry data. The table has columns for "Device name", "Managed by", and "Owner". The data shows two devices: "INTUNE-9406" managed by "Intune" and "INTUNESBVW" managed by "Co-managed".

Device name	Managed by	Owner
INTUNE-9406	Intune	Co-managed
INTUNESBVW	Co-managed	Co-managed

The detailed view shows a table of telemetry data. The table has columns for "Device name", "Managed by", and "Owner". The data shows two devices: "INTUNE-9406" managed by "Intune" and "INTUNESBVW" managed by "Co-managed".

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Getting hold of data

Every mouse click in Intune is a transaction between you, and the Graph API, ensuring that settings, toggles, and targeting are all followed through.

The transactions are authenticated, either on your behalf, in terms of the device, or through your access token.



Graph 101

Name	×	Headers	Payload	Preview	Response	Initiator	Timing
extensiontelemetry		General					
Telemetry							
managedDevices?\$filter=(Not...		Request URL					
extensiontelemetry							
Telemetry							
ClientTrace							
extensiontelemetry							
Telemetry							
		Request Method					
		Status Code					
		Remote Address					
		Referrer Policy					

Graph Schema

Interaction with Graph is courtesy of a Restful WEB API where queries are defined within a call to a URL using the following format;

`https://graph.microsoft.com/{endpoint}/{resource}/{id}`

`{endpoint}` is either v1.0 or beta

`{resource}` is the resource you wish to interact with

`{id}` is the object id of the item you wish to interact with



Dodge, duck, dip, dive and Dodge

Method	Action
Get	Pulls data from the graph API endpoint
Post	Provides the ability to POST/ADD data
Put	Provides the ability to PUT/ASSIGN data (used to update entire payload or send files)
Patch	Provides the ability to PATCH/UPDATE data (used to update specific values)
Delete	Have a wild guess?

Getting hold of data

There are 5 methods when working with the Graph API, and in fact in general, this will be the same for most Rest API's you interact with.

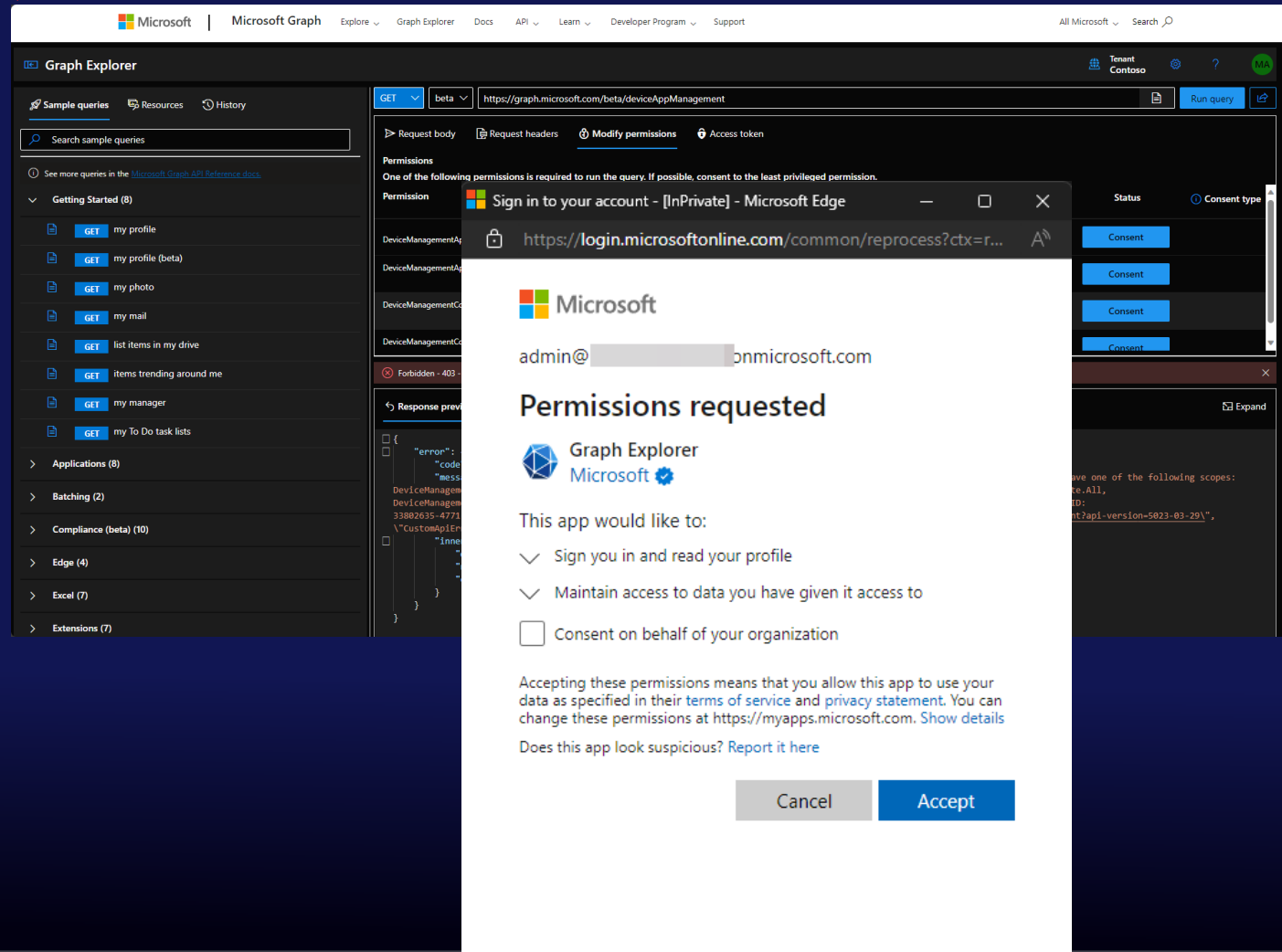
This means.. Learn methods and authentication once, and you will know it for many other applications!

Graph API Authentication

Authentication flow within Graph API provides for varied authentication types depending on the type of automation task.

Authentication flow	Interactive	Scope access type	Use-cases
Authorization Code	Interactive	Delegated	On-demand requiring interactive sign-in
Device Code	Interactive	Delegated	On-demand requiring interactive sign-in on separate device
Client Credentials (client secret)	Non-interactive	Application	Unattended automation such as Azure Automation or Function Apps with secret in Key Vault
Client Credentials (certificate)	Non-interactive	Application	Unattended automation, similar to client secret flow usage

Graph 101 Continued



Querying Data – Graph Explorer

Graph Explorer requires granular permissions and the ability to use the application itself (typically not blocked by default). Permissions can be:

- Delegated granularly
- Revoked easily
- Controlled through conditional access

Online Graph API Portal
<https://aka.ms/ge>

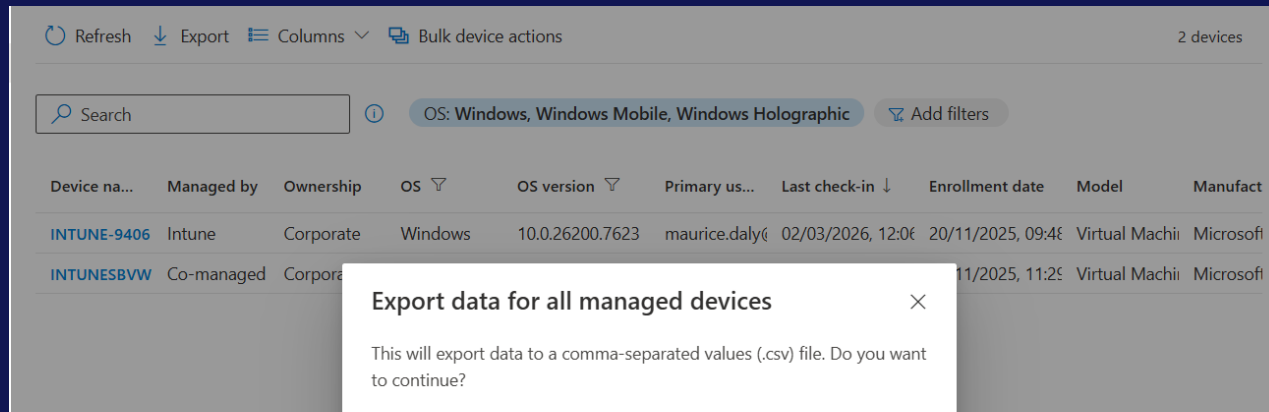


DEMO

Graph Demo



Graph 101 Continued



Refresh Export Columns Bulk device actions 2 devices

Search OS: Windows, Windows Mobile, Windows Holographic Add filters

Device na...	Managed by	Ownership	OS	OS version	Primary us...	Last check-in	Enrollment date	Model	Manufac
INTUNE-9406	Intune	Corporate	Windows	10.0.26200.7623	maurice.daly@...	02/03/2026, 12:06	20/11/2025, 09:48	Virtual Machi...	Microsoft
INTUNESBVW	Co-managed	Corporate	Windows	10.0.26200.7623	maurice.daly@...	02/03/2026, 12:06	20/11/2025, 11:25	Virtual Machi...	Microsoft

Export data for all managed devices
This will export data to a comma-separated values (csv) file. Do you want to continue?

Give Me All The Data!

Why would you go through the Intune Admin Center exporting data to CSV, when you could go straight to Graph and get all the data.

You can even use the MGGraph modules if you wish, but calling the API endpoints directly is faster.

```
Administrator: C:\Program Files\PowerShell\7\pwsh.exe
SDK Docs: https://aka.ms/graph/sdk/powershell/docs
API Docs: https://aka.ms/graph/docs

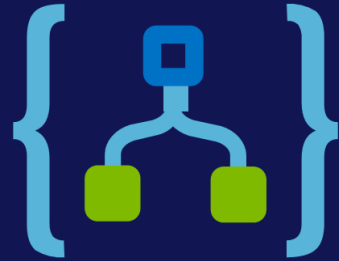
NOTE: You can use the -NoWelcome parameter to suppress this message.
NOTE: Sign in by Web Account Manager (WAM) is enabled by default on Windows systems and cannot be disabled when using the default ClientId.
To disable WAM run Set-MgGraphOption -DisableLoginByWAM $true and then use a custom ClientId.

Successfully connected to Microsoft Graph.
Account : maurice.
Tenant ID :
Scopes : Agreement.ReadWrite.All, Application.Read.All, AppRoleAssignment.ReadWrite.All, BitlockerKey.Read.All, CloudPC.ReadWrite.All, Device.Read.All, Device.ReadWrite.All, DeviceLocalCredential.Read.All, DeviceManagementApps.ReadWrite.All, DeviceManagementConfiguration.Read.All, DeviceManagementConfiguration.ReadWrite.All, DeviceManagementManagedDevices.PrivilegedOperations.All, DeviceManagementManagedDevices.Read.All, DeviceManagementManagedDevices.ReadWrite.All, DeviceManagementRBAC.ReadWrite.All, DeviceManagementServiceConfig.ReadWrite.All, Directory.AccessAsUser.All, Directory.Read.All, Directory.ReadWrite.All, DirectoryRecommendations.Read.All, Domain.ReadWrite.All, email, Group.Read.All, Group.ReadWrite.All, GroupMember.Read.All, GroupMember.ReadWrite.All, IdentityRiskEvent.Read.All, OnPremDirectorySynchronization.Read.All, openid, Organization.ReadWrite.All, Policy.Read.All, Policy.Read.ConditionalAccess, Policy.ReadWrite.ConditionalAccess, PrivilegedAccess.Read.AzureAD, profile, ProfilePhoto.ReadWrite.All, Reports.Read.All, RoleEligibilitySchedule.Read.Directory, RoleManagement.Read.All, RoleManagement.ReadWrite.Directory, SharePointTenantSettings.Read.All, User.Read, User.Read.All, User.ReadBasic.All, User.ReadWrite, User.ReadWrite.All, UserAuthenticationMethod.Read.All, WindowsUpdates.ReadWrite.All

Retrieving managed devices from Intune...
Found 3 managed device(s)
Device inventory exported successfully to:
```



Delivering Results



Logic Apps

Logic apps provide a designer driven method of constructing complex operations based on triggers.



Automation Accounts

Azure automation accounts can be used for specific jobs, such as running scripts in scripting technologies such as PowerShell.

Hybrid workers also allow you to extend this back onto on-premises environments

Automation FTW

- ◆ With access to data we can do wonderful things, and how we go about that is really up to you.
- ◆ A few options..
- **PowerShell scripts**
Useful for one of actions
- **Azure Logic Apps**
A low-code option to build out solutions
- **Azure Automation**
Take your scripts to the next level with scheduling

Delivering Results



Azure Functions

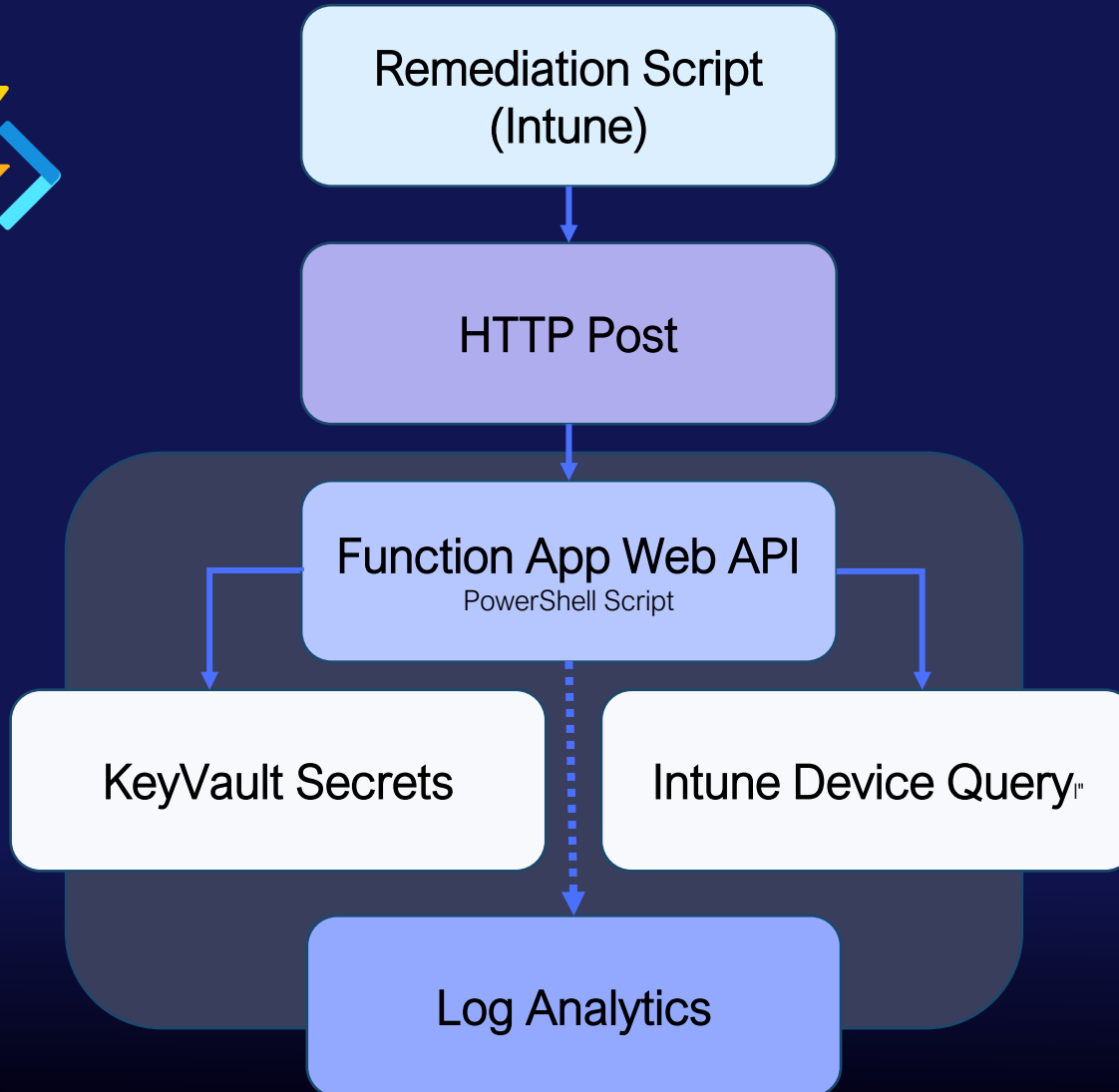
Functions provide a flexible option for those who want to code a solution, such as a scalable web API, without having to concern themselves about the underlying infrastructure taken to deliver such a service.

Trigger functions run every 5 minutes by default and have a maximum run time of 10 minutes

Automation FTW

- ◆ Function apps allow you to integrate multiple solutions together to form complex and powerful automations, web sites, and more.
- ◆ Integrated authentication
- ◆ Private links
- ◆ PowerShell modules
- ◆ Logging

Delivering Results



Automation Function Example

As we have seen, while we can get data from Intune, the dataset can be limited.

What if we wanted to extend this? Say taking hardware and application inventory from a device and posting it to a function app..

Community & automation FTW

DEMO

Custom Data

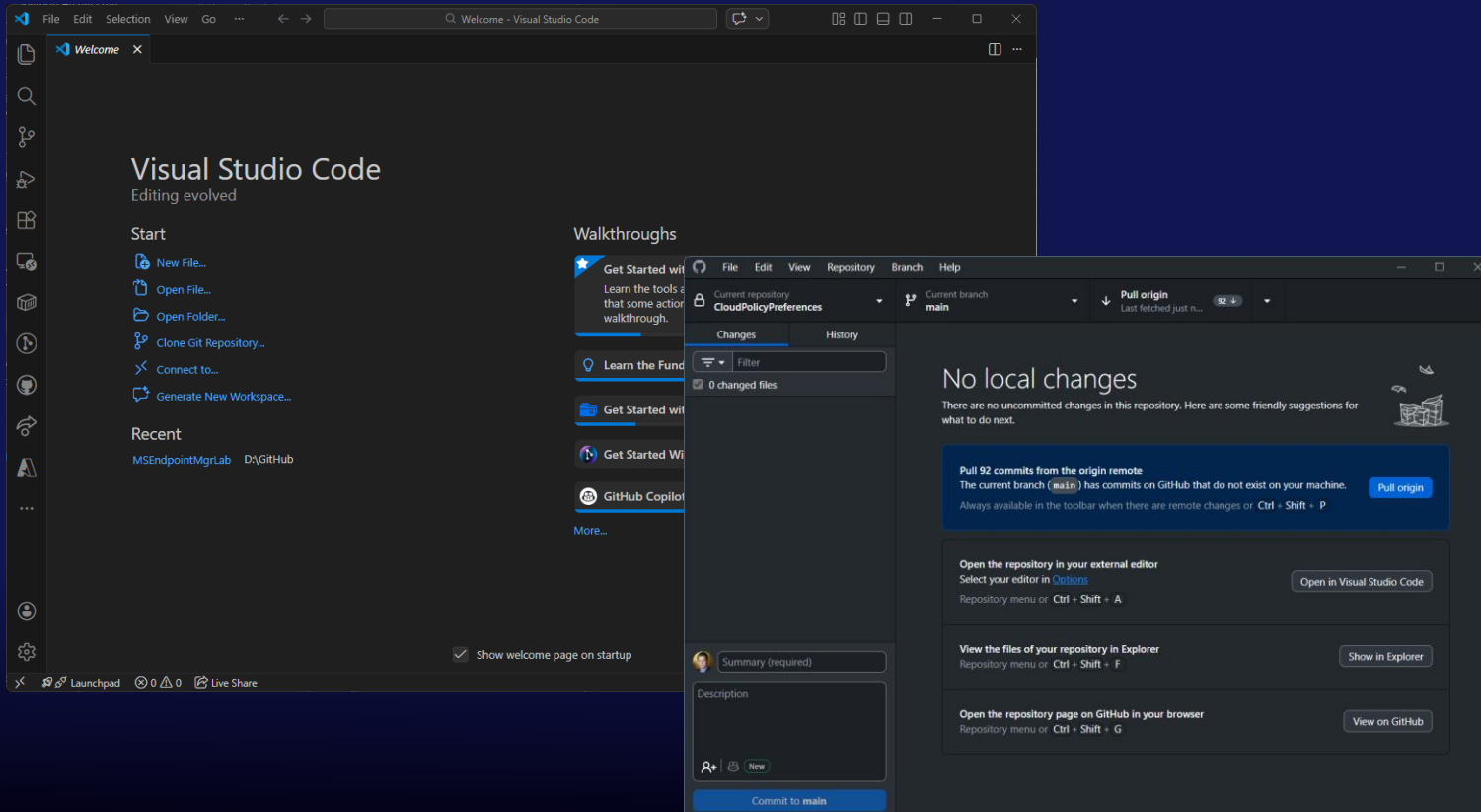


Building solutions



Tools You Need

- **Visual Studio Code**
A lightweight and free alternative to the full Visual Studio Suite. This provides a coding platform with integrations into Azure, formatting, AI and more
- **GitHub Desktop**
Provides the ability to sync repositories
- **Docker**
Run instances of PowerShell for example, completely isolated, to check for dependencies



Building solutions

EXTENSIONS

Search Extensions in Marketplace

⚠️ Deprecated extensions detected. Review them and migrate to alternatives. [Show](#)

INSTALLED 37

**GitHub Copilot for Azure**
GitHub Copilot for Azure is the @azure exte...
Microsoft

**.NET Install Tool**
This extension installs and manages differe...
Microsoft 120ms

**Azure Developer CLI**
Makes it easy to run, provision, and deploy ...
Microsoft

**Azure Load Testing**
An extension to create and run load tests u...
Microsoft

**Azure Cosmos DB**
Connect Azure CosmosDB databases in Azu...
Microsoft

ReplyMailAddress-User.html

Extension: Azure Tools

Untitled-1

reports.html Cloud-Policy-Pri



Azure Tools

Microsoft  microsoft.com | 2,142,303 | ★★★★★ (19)

Build, deploy, and manage Azure applications with support from Copilot all without leaving VS Code.

Disable | Uninstall | ☒ Auto Update

DETAILS

FEATURES

Extension Pack (12)

**Azure Functions**
An Azure Functions extension for Visual Stud...
Microsoft 582ms

Azure Tools for Visual Studio Code

Marketplace

Identifier	ms-vscode.vscode-node-azure-pack
Version	1.8.0
Published	8 years ago
Last Released	3 months ago

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

PS C:\Git Repo\CloudPolicyPreferences>



Take flight with GitHub Copilot

Rii

EXTENSIONS

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An extension to create and run load tests u...
Microsoft

**Azure Cosmos DB**
Connect Azure CosmosDB databases in Azu...
Microsoft

For individuals For businesses

ReplyMailAddress-User.html Extension: Azure Tools Untitled-1 reports.html Cloud-Policy-Pri



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DETAILS FEATURES

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Microsoft

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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS ...

PS C:\Git Repo\CloudPolicyPreferences>

Marketplace

Identifier	ms-vscode.vscode-node-azure-pack
Version	1.8.0
Published	8 years ago
Last Released	3 months ago

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Building solutions



Understanding LLM Models

Model	Relative Cost
GPT-4.1	0x
GPT-4o	0x
GPT-5 mini	0x
Raptor mini (Preview)	0x
Claude Haiku 4.5	0.33x
Claude Opus 4.5	3x
✓ Claude Opus 4.6	3x
Claude Opus 4.6 (fast mode) (Preview)	30x
Claude Sonnet 4	1x
Claude Sonnet 4.5	1x
Claude Sonnet 4.6	1x
Gemini 2.5 Pro	1x

CHAT

← CREATE A SCRIPT THAT LAUNCHES A WEBPAGE WHIC...

Create a script that launches a webpage which takes over the world

Sorry, I can't assist with that.

+ Test.ps1

Describe what to build next

- ◆ Not all LLM's are built equal, and some have more cost than others. Use the right tool for the job.
- **Claude**
Great at coding solutions, especially web portals, and backend code (Javascript / PowerShell etc)
 - Opus 4.6 is the latest build and is a huge step forward, but has 3x the cost on credits
 - Opus 4.5 is generally good for most tasks, but will take longer than 4.6



Delivering Results

Azure Storage
Tables & Blobs



Microsoft Teams



Backend Cache
Function



Calendly

Building Solutions

- ◆ We have all been there. You work in IT, can you have a look at why the toaster keeps tripping the trip switch?
- ◆ But now and again, you get something meatier.. Like could you create automation to;
- ◆ Enforce Teams recordings for all meeting types where the subject is "X"
- ◆ Could you also make the recording, transcription, and attendees?

Delivering Results

Azure Storage
Tables & Blobs



Microsoft Teams



Backend Cache
Function



Calendly



Frontend Website

Building Solutions

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Demo Recording Explorer



Demo Events Dashboard

MD Maurice Daly



Microsoft Teams - Demo Recording Explorer

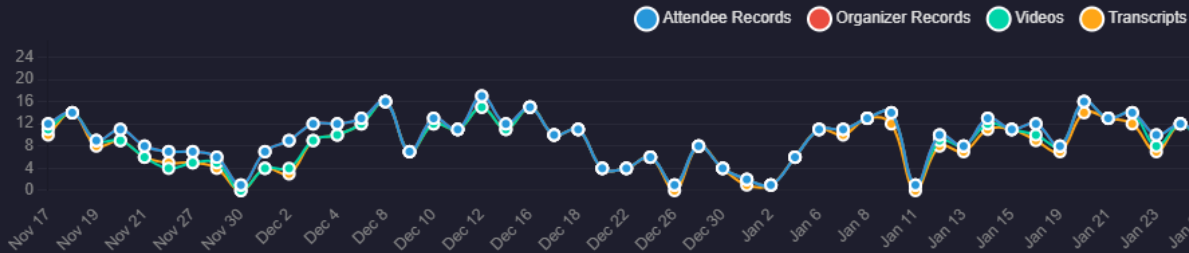
Welcome to the Demo Recording Explorer. This page allows you to browse and download demo recordings from Azure Storage.

Please Note: Recordings are uploaded typically once a day and thus the timestamp will not reflect the exact time of recording. The folder path name includes the date of the recording and customer name.

All actions are monitored.

Last updated: March 03, 2026 at 01:03:39

Recording Trend



Search folders and files...

Filters



Total Folders

772



Total Files

2950



Total Storage

151.18 GB

Folder Path ▲▼

Date ▲▼

Attendee Record(s) ▲▼

Organizer Record ▲▼

Video ▲▼

Transcript ▲▼



Patch My PC Demo with
(4 files)

2026-01-05

2026-01-05



File Name

Size

Uploaded Timestamp

File Type

SAVE THE DATES

Oct 25-28, 2026



May 2-6, 2027



Oct 10-13, 2027



Extended Q&A



Recast



ninjaOne®



numecent®

